Rotor Shaft Turning on Machining Centre MC-500

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Document type	Work Instruction	**Status**	Current
Effective date	2025-11-15	**Owner**	Production Engineering
Classification	Internal		

1 Purpose

This instruction covers the turning of rotor shafts RS-40 on the machining centres MC-500 in hall 3. It describes the checks the operator has to do during the shift.

2 Safety

Never open the machine door while the spindle is turning. Remove chips only with the chip hook, never by hand. Wear safety glasses at all times in hall 3.

3 Coolant

The coolant concentration must be between 6 % and 8 %. Check it with the refractometer twice per shift, at shift start and after the break. If the value is below 6 %, add concentrate according to the mixing table at the machine. If it is above 8 %, add water. Record every check on the machine log sheet.

4 Tool change

Replace the finishing insert after 450 parts or earlier if the surface roughness of the bearing seat is above Ra 0.8 um. After every insert change, measure the first part completely (first-part inspection) before releasing the machine.

5 In-process measurement

Characteristic	Specification	Frequency	Gauge
Bearing seat diameter	40 k6 (+0.018 / +0.002 mm)	every 25th part	air gauge AG-40
Surface roughness bearing seat	Ra max 0.8 um	every 50th part	roughness tester
Overall length	312.0 +/- 0.1 mm	every 50th part	height gauge
Runout	max 0.02 mm	every 50th part	dial indicator

If a measured value is outside the specification, stop the machine, segregate all parts produced since the last good measurement and inform the shift leader and quality.

6 Warm-up

After a standstill of more than 8 hours, run the spindle warm-up program before production (see MM-CNC-003).

7 Revision history

Rev	Date	Change
A	2025-11-15	First issue